SynapseLex: End-to-End Axiomatic Neuro-Symbolic Legal Adjudication

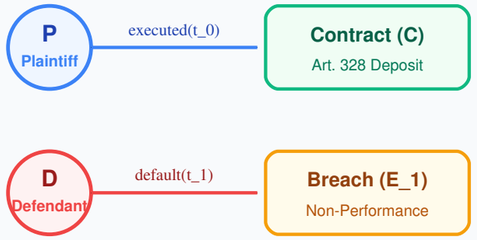
G(t) → TGAT(Φ_d) → TBox(M_proj) → GE(AF)

STAGE 1 Dynamic Legal Graph

G(t) = (V, E, R, T, S) | Dual-Clock Continuum

Bipartite Event Graph

|V|=4, |E|=3



Dual-Clock Deontic Axis

τ(t) ≠ t



$$\tau(t) = \int [\, 1 - 1_tolling(\sigma) \,]\, d\sigma$$

$$\Delta\tau = \tau(t_suit) - \tau(t_1)$$

- Tolling Interval: T_freeze = 123d
- Prescription Threshold: 1,095d (3y)

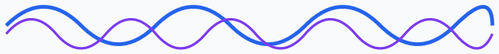
Δτ = 1,254d > 1,095d → Art. 429 Lapsed

STAGE 2 Axiom-TGAT Attention

Continuous Harmonic Spectral Embeddings

Bochner Harmonic Basis

Φ_d(Δτ) ∈ R^32



$$\Phi_d(\Delta\tau) = \sqrt{(2/d)}\, [\cos(\omega_k\, \Delta\tau) \,\|\, \sin(\omega_k\, \Delta\tau)]$$

$$\alpha_uv = \text{Softmax}(\, (q_u^T\, k_v) / \sqrt{d_k}\,)$$

$$k_uv = [\, h_v \,\|\, e_uv \,\|\, \Phi_d(\Delta\tau)\,] \in R^{(2d + d_\tau)}$$

$$m_u = \oplus_h \Sigma_v\, \alpha_uv^{(h)}\, v_uv^{(h)} \in R^d$$

Hyperparams: H=4 heads | d=128 | d_τ=32

Contract Mutation Recurrence

s^(t) ∈ Δ^3

$$s^{(t+1)} = \sigma(\, W_s\, s^{(t)} + U_s\, h(t) + b_s\,)$$

- Banach Factor: L = 1/4 ||W_s||_2 < 1
- **Guarantees Unique Fixed Point Equilibrium s***

Predicted State Distribution P(S):

S_0 (Valid): 0.02% | S_1 (Voidable): 0.11%

S_2 (Void Ab Initio): 0.01%

S_3 (Extinguished Action): 99.86%

STAGE 3 Ontological Arbiter

VCLO Description Logic TBox Loss

VCLO TBox Axioms (SROIQ)

233 Axioms

Art. 129(2)	Perf ≥ 2/3 ⊑ CuredValid
Art. 429	Δτ > 1095d ⊑ DismissBarred
Art. 156	ForceMajeure ⊑ dτ/dt = 0

HermiT DL Tableau Audit

Satisfiability Verified: TBox ⊭ ⊥ (0.42s)

Differentiable Axiomatic Loss

LCR = 99.87%

$$L_total = L_task + \lambda_1\, L_TBox + \lambda_2\, L_state$$

$$L_TBox = \Sigma_k \max(0, \psi_k(y_hat, G))$$

λ_1 = 2.0 | λ_2 = 1.0 (AdamW, η=10^-3)

Theorem 1 Analytic Convergence:

$$\lim_{T \to \infty}\, 1/T\, \Sigma\, L_onto \;\le\; O(\, 1 / (\lambda\, \sqrt{T})\,)$$

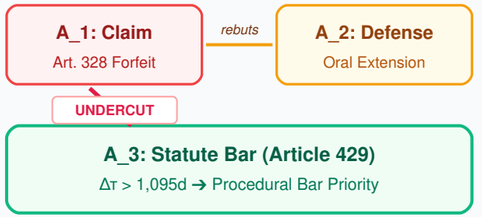
- Amortized Matrix M ∈ {-∞, 0}^(4×4)
- **Inference in O(1) time (8.15 ms vs 2.8s LLM)**
- **0.13% TOVR (Zero-Violation Compliance)**

STAGE 4 Defeasible Head

Phan Minh Dung Abstract Argumentation

Dung Argument Graph AF

GE = [A_3]



$$F_AF(\varnothing) = [A_3] \rightarrow GE(AF) = [A_3]$$

Knaster-Tarski Least Fixed Point Semantics

Judicial Decree Adjudication

$$z_masked = z + m_proj(3)$$

m_proj(3) = (-∞, -∞, 0, -∞)^T

$$P(\text{Forfeit} \mid \Delta\tau > 1095) \equiv 0$$

Strict 100% Soundness under Law

AUDITED VERDICT: DISMISSED (CODE 2)

Statute-Barred Procedural Preemption

90.17% Hits@1 | 100% Statute Rec.